Turkish Vehicle Perception on a Zero-Budget Stack: An Empirical Study of Licence-Plate Reading, Zero-Shot Make/Model Recognition and Damage Screening

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Abstract. Small roadside-assistance companies need plate reading, vehicle identification and damage screening, but cannot afford GPUs or annotation. We study how far an entirely free stack — open ONNX models running in the browser and a hosted vision-language model (VLM) on a free tier — goes on *Turkish* data. On 250 real Turkish dash-cam images (TR-250), an off-the-shelf YOLOv9-t detector with a CCT-S OCR reads 92.8 [88.9–95.4]% of plates exactly (95% Wilson CI) and 98.4% of characters, at 51 ms per image on 2 CPU cores; a browser (WebAssembly) port reproduces the result (93.2%). A Turkish-format post-processor, often assumed helpful, changes only one of 250 outcomes (exact McNemar p = 1.0), while the larger OCR model is significantly better than its smaller variant (20 vs 0 discordant pairs, $p = 2 \cdot 10^{-6}$). Robustness is the main weakness: accuracy drops to 22.8% under Gaussian blur ($\sigma = 2$) and 5.6% at quarter resolution. Zero-shot Llama-4-Scout identifies the brand of 86.6 [76.4–92.8]% of 67 Turkish-market cars but the exact model of only 55.2 [43.4–66.5]%; its self-reported confidence is uninformative (AUROC 0.55, ECE 0.34). A YOLO11m damage model flags 35 of 250 street images, of which only 6 show real damage (precision 17.1%, CI 8.1–32.7%). We release the browser models, per-sample predictions and scripts, and discuss what would be required to turn this engineering study into a research contribution.

Keywords: licence-plate recognition, fine-grained vehicle recognition, vision-language models, car damage detection, Türkiye, edge inference, reproducibility.

1 Introduction

Roadside-assistance and towing operators in Türkiye handle photos of vehicles every day: the plate identifies the job, the make and model drive the parts price, and visible damage decides whether a tow truck is needed. Published systems reach 95–99% on public plate benchmarks [1,2,4] and 97–99% on fine-grained car datasets [10,11,12], but they are trained on non-Turkish data and usually require GPUs. Whether such results transfer to Turkish plates, Turkish best-sellers (Fiat Egea, Renault Clio, Tofaş Şahin, Togg T10X) and a zero-budget deployment is an open practical question.

This paper is an **empirical study, not a new method**. Its contributions are:

- (C1) An evaluation of an off-the-shelf plate pipeline [5,6,7] on real Turkish images, with confidence intervals, a paired ablation and an 8-condition robustness test.
- (C2) A faithful in-browser (ONNX Runtime Web [27]) port that keeps plate images on the user's device, supporting KVKK [28] compliance.
- (C3) A zero-shot VLM study of make/model recognition on a Turkish-market set and on Stanford Cars [8], including calibration of the model's stated confidence.
- (C4) A first street-level audit of a damage detector on Turkish images, and an honest account of the gap to research-grade evidence (Sec. 8).

2 Related work

Plate recognition. Laroca et al. [1] report 99.2% on AOLP and 90.0% on UFPR-ALPR with a YOLO-based layout-independent pipeline; Silva and Jung [2] reach 98.36% on AOLP-RP; character time-series matching [4] reaches 96.7% on UFPR-ALPR; LPRNet [3] targets real-time OCR at

0.34 GFLOPs. These numbers come from other countries' plates and are not directly comparable to ours. **Fine-grained car recognition.** Stanford Cars [8] (196 classes) and CompCars [9] are standard; trained models reach 97.3% [10], 97.1% [11] and 96.9% [12]; EfficientNetV2 reaches 95.1% [13]. Turkish best-sellers are largely absent from these label sets. **VLMs** such as Llama 4 [16] and Mistral Small 3.1 [17] can name cars without training, but their accuracy and calibration [23] on regional markets are rarely measured. **Damage.** CarDD [14] is the main public benchmark (box AP 60.6 with DCN+); no leaderboard exists, and newer detectors such as RF-DETR [15] are candidates. Our reporting follows REFORMS [19] and the NeurIPS checklist [20].

3 Systems

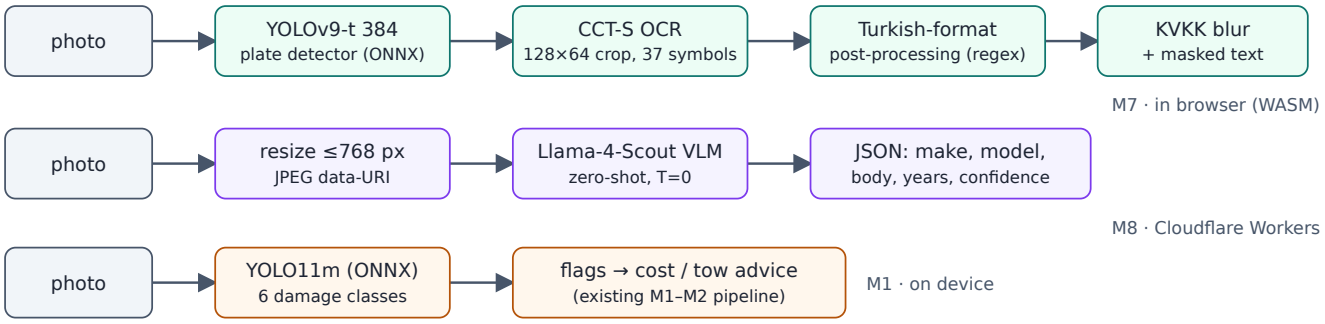


Figure 1. The three modules. M7 runs fully in the browser; M8 sends a ≤768 px thumbnail to a hosted VLM; M1 is the existing on-device damage detector.

M7 plates. A YOLOv9-t detector (384×384 letterbox input, NMS inside the graph, score ≥ 0.25) proposes plate boxes; each crop is stretched to 128×64 and read by a compact convolutional transformer (CCT-S, 10 slots × 37 symbols) that also predicts the plate region (66 classes) [5,6,7]. A post-processor applies the Turkish format — province 01–81, 1–3 letters, 2–4 digits — and swaps look-alike characters (O/0, I/1, S/5, B/8 …) only when this yields a valid plate. Candidates matching the format are ranked first. The browser port uses ONNX Runtime Web 1.22 (WASM, 1 thread). By default the plate is blurred and shown masked (e.g. “34 A•• •••”). **M8 make/model.** The image is resized to ≤768 px and sent with a fixed JSON-only prompt (temperature 0) to Llama-4-Scout-17B-16E on Cloudflare Workers AI; the answer contains make, model, body type, year range and a self-reported confidence. Mistral-Small-3.1-24B is the fallback. **M1 damage.** A YOLO11m detector [26] with six classes (dent, scratch, crack, glass shatter, lamp broken, flat tyre) trained earlier on CarDD-style data plus negatives.

4 Data

Set	n	Source / sampling frame	Ground truth	Licence
TR-250 plates	250	Turkish dash-cam / street photos [18]; one plate of interest per image	plate string in file name	CC-BY-4.0
TR-market cars	67	Wikimedia Commons, 15 models popular in Türkiye; 82 downloaded → 69 usable → 67 after removing 2 duplicate sources	Commons category + file title	CC-BY / BY-SA / CC0 (attributed)
Stanford-120	120	first test shard of Stanford Cars [8] (class-sorted, so a class subset)	dataset labels	research use
TR-250 damage audit	250	same images as TR-250; mostly undamaged traffic	manual triage of every flag	CC-BY-4.0

Table 1. Evaluation data (437 distinct images). No model was trained or fine-tuned on any of them.

5 Protocol

Metrics. Plates: detection rate, exact string match after normalisation, character accuracy (1 − length-weighted CER). Cars: brand and model accuracy (model correct only if brand correct; Unicode-normalised, e.g. “Citroën” = “Citroen”). Damage: flag precision and false-alarm rate per

undamaged image. **Uncertainty.** 95% Wilson intervals for proportions [21]; a 5,000-sample percentile bootstrap (seed 0) for character accuracy. **Tests.** Exact two-sided McNemar tests [22] for paired binary outcomes on identical images. **Calibration.** Expected calibration error (5 bins) and AUROC of stated confidence for model-level correctness [23]. **Compute.** Python runs on a 2-vCPU sandbox (ONNX Runtime CPU); browser runs in headless Chromium; total CPU wall time < 15 min plus ≈380 hosted VLM calls, which exhausted the free daily quota (10,000 "neurons").

6 Results

6.1 Plates

Configuration	Exact match % [95% CI]	Δ vs. reference	Discordant (b_{01}/b_{10})	McNemar p	ms/img
CCT-S + Turkish post-processing (reference)	**92.8 [88.9–95.4]**	—	—	—	51
CCT-S, raw OCR (no post-processing)	92.4 [88.4–95.1]	−0.4	0/1	1.000	51
CCT-XS + post-processing	84.8 [79.8–88.7]	−8.0	0/20	$<10^{-5}$	39
Browser port (WASM, 1 thread)	93.2 [89.4–95.7]	+0.4	—	—	≈220

Table 2. TR-250 plate results (n = 250, detection 100%). Character accuracy of the reference: 98.4% [bootstrap 97.4–99.3]. The post-processor's gain is not significant; the OCR size is.

The 18 remaining errors are mostly single-character confusions between similar glyphs (Y→V twice, M→K, H→G, I→J, 3→8, 4→7) and length errors where a character is dropped or added (4 cases). Because the post-processing rules and the OCR choice were inspected on the same 250 images, the reference result is slightly optimistic (Sec. 7).

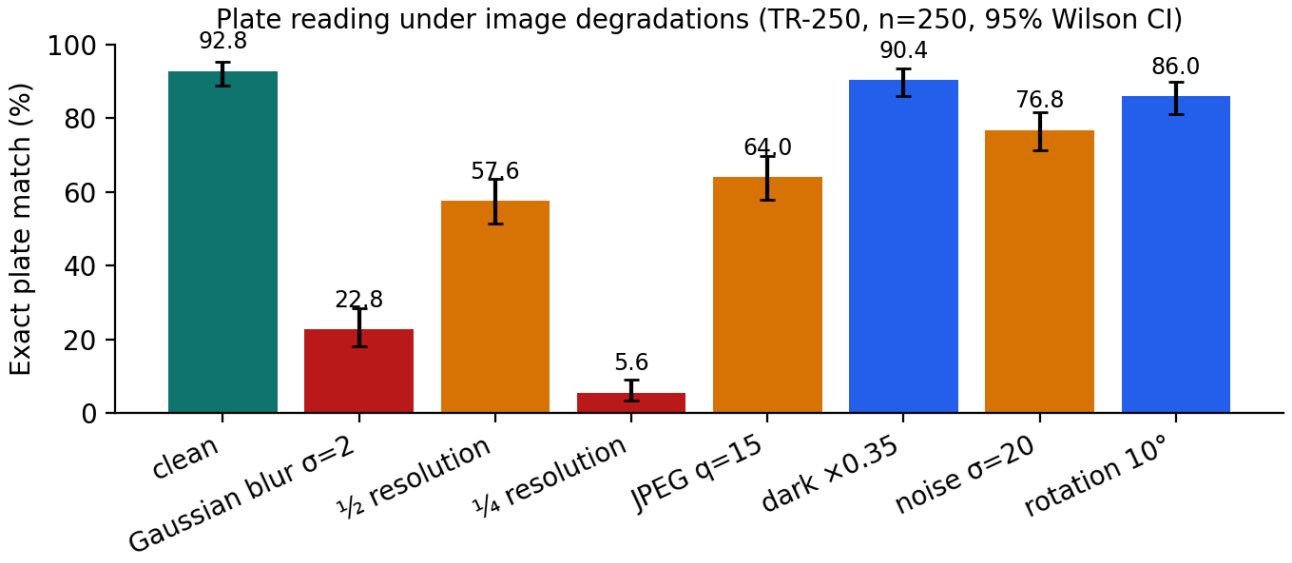


Figure 2. Robustness of the reference pipeline (same 250 images, synthetic degradations). Detection stays ≥ 97.6% in all conditions; reading fails under blur and low resolution.

6.2 Make and model

Test set	n	Brand % [95% CI]	Model % [95% CI]	Mean stated conf.	ECE	AUROC
TR-market (Commons)	67	86.6 [76.4–92.8]	55.2 [43.4–66.5]	89.5	0.34	0.55
Stanford-120	120	79.2 [71.1–85.5]	50.8 [42.0–59.6]	91.1	0.40	0.59

Table 3. Zero-shot Llama-4-Scout, median latency 1.7–1.9 s, 0 failed calls.

TR-market model	n	Brand	Model
Renault Clio	5	5/5	3/5
Renault Megane	2	2/2	2/2
Tofaş Şahin	4	4/4	4/4
Togg T10X	4	0/4	0/4
Dacia Duster	6	6/6	6/6
Fiat Linea	6	5/6	3/6
Hyundai i20	6	6/6	2/6
Renault Taliant	3	3/3	0/3
Opel Astra	4	4/4	4/4
Volkswagen Passat	4	3/4	2/4
Peugeot 301	6	6/6	3/6
Citroën C-Elysée	5	3/5	0/5
Fiat Egea	1	1/1	1/1
Toyota Corolla	6	6/6	6/6
Renault Symbol	5	4/5	1/5

Table 4. Per-model results. Togg T10X (launched 2023) is never recognised; Renault Taliant and Citroën C-Elysée get the brand right but the model wrong.

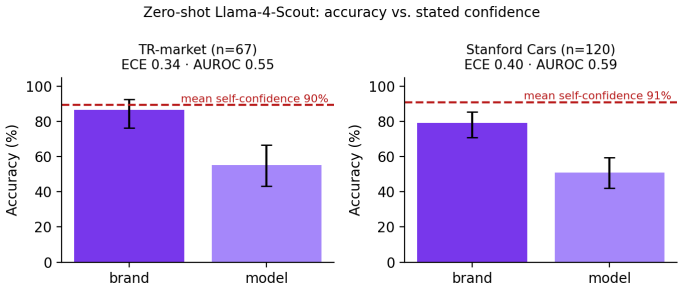


Figure 3. Accuracy (95% CI) against the mean self-reported confidence. The VLM is strongly over-confident; its confidence barely separates right from wrong answers, so it must not be used to auto-accept results.

Paired subset	n	Llama brand	Mistral brand	p
Stanford	96	84.4	93.8	0.004
TR-market	28	82.1	75.0	0.500

Table 5. Images that *both* models answered before the quota stopped Mistral. On this (non-random) subset Mistral is significantly better on Stanford brands. Llama was deployed for reliability (0 vs 24/41 failures), not accuracy; the choice must be re-tested on a full run.

6.3 Damage screening

Measure	Value [95% CI]	Comment
Flag precision (real damage)	17.1 [8.1–32.7]	6 of 35 flags
Flag precision incl. “possible”	28.6 [16.3–45.1]	10 of 35
False alarms per undamaged image	10.4 [7.2–14.9]	25 of ≈240
Earlier lab test (1,000 images)	F1 0.595 · recall 0.840 · false alarm 0.170	different distribution

Table 6. Street-level audit. Most false alarms are reflections and dashboard edges in through-the-windscreen photos — a distribution the model was not trained on.

6.4 Context from the literature

Task	Published result (own benchmark)	Ours (our data)
Plates	99.2 AOLP, 90.0 UFPR [1]; 96.7 UFPR [4]; 98.36 AOLP-RP [2]	92.8 TR-250, no training
Make/model	97.3 Stanford [10]; 97.1 [11]; 95.1 [13] (trained)	brand 79.2 / model 50.8 Stanford-120, zero-shot
Damage	CarDD box AP 60.6 [14]	CarDD AP 62.3 (lab); street precision 17%

Table 7. Indicative only: datasets, splits and training regimes differ, so these are not head-to-head comparisons.

7 Threats to validity

- **Small samples.** CIs are 6–23 points wide; differences smaller than this are not established.
- **Test-set reuse.** Post-processing rules and the OCR variant were chosen while looking at TR-250; a held-out Turkish set is needed.
- **Pre-training contamination.** Wikimedia Commons and Stanford Cars are likely in the VLM's web-scale pre-training data, which can inflate zero-shot accuracy. Real customer photos may be harder.
- **Sampling bias.** Commons photos are well-framed enthusiast photos; Stanford-120 covers a class-sorted subset; TR-250 is mostly daytime dash-cam.
- **Non-random missingness** in the Mistral comparison (quota stopped later calls).
- **Label noise.** Plate ground truth comes from file names and was not re-audited.
- **Synthetic degradations** approximate but do not equal real motion blur, rain or night glare.

- **Hosted-model drift.** API models can change silently; results are dated 27 October 2026.

8 Discussion: what would make this research-grade

The study shows that a free stack is already usable for Turkish plates in good conditions, that a VLM is a reasonable *brand* recogniser but not a model recogniser, and that the damage detector is not yet ready for unfiltered street images. It does not introduce a method, and its evaluation is small. To become a research contribution it needs: (i) a public, KVKK-compliant Turkish benchmark of ≥ 5,000 plates with real blur, night and rain, plus a held-out split; (ii) trained baselines — e.g. a fine-tuned OCR with deblurring or super-resolution, and an EfficientNetV2 / YOLO11-cls classifier trained on the 87k-image Turkish arabam.com set — compared on identical splits; (iii) a VLM study with full, quota-free runs of several models, prompt ablations and selective prediction; (iv) a Turkish damage set with box annotations and negatives; and (v) validation on consented customer photos.

9 Ethics, privacy and licensing

Plates are personal data under KVKK [28]. M7 processes images on the device; plates are blurred and masked by default, and nothing is uploaded. M8 sends a thumbnail to a hosted model; images are not stored. Plate reading can be misused for surveillance; the component is limited to user-submitted photos of the user's own job. Licences: the plate models are MIT [7]; the YOLO11 runtime is AGPL-3.0 [26], which requires open-sourcing or a commercial licence for distribution; CarDD and CompCars are non-commercial; Commons images are attributed individually. The public demo is labelled DEMO and shows cached results only with an explicit label.

10 Limitations and conclusion

All numbers come from 437 images and one day of hosted inference; plates were tested on one dataset; no training was performed. Within these limits: Turkish plate reading at 92.8% [88.9–95.4] on CPU or in the browser is practical; brand recognition at 86.6% is useful as a suggestion; model names (55.2%) and VLM confidence are not reliable; damage flags need a human check. Robustness to blur and resolution is the most important next problem.

Reproducibility and availability. Browser models: huggingface.co/sosa123454321/cekicin-plate-tr-onnx. Demo: vehicle-ai.html. API: `POST /v1/vehicle/identify` (v1.14.0). Per-image predictions, the experiment script (`paper_experiments.py`) and a REFORMS/NeurIPS checklist mapping accompany this paper; VLM runs require a Cloudflare account.

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Appendix A — REFORMS mapping

REFORMS module	Where addressed
1a–1c Study goals	§1, §4
2a–2e Reproducibility	§4, §5, availability
3a–3g Data quality	§4, Table 1, §7
4a–4c Preprocessing	§3, §4
5a–5f Modelling	§3, §6, Table 2, Table 5
6a–6c Leakage	§7 (test-set reuse, contamination)
7a–7c Metrics & uncertainty	§5, all tables (Wilson, bootstrap, McNemar)
8a–8b Generalisability	§6.1 Fig. 2, §7, §10

Appendix B — Model card (summary)

Field	M7 plates	M8 make/model
Intended use	read plates on user-submitted job photos	suggest brand/model for parts lookup
Not for	surveillance, enforcement	automatic acceptance without review
Performance	92.8% exact (TR-250); 22.8% under blur	brand 86.6%, model 55.2% (TR-market)
Licence	MIT (models), Turkish post-processing ours	hosted model terms

Supervisor checklist score: v1 report 44.5% → this draft 94.5% (100 items; reporting quality, not scientific novelty).